

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1110

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing techniques. (BL4,BL5)
Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:40

Annexure A

MOCK INTERVIEW

Code of Conduct:

I, Tejaswini.M(192411079), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

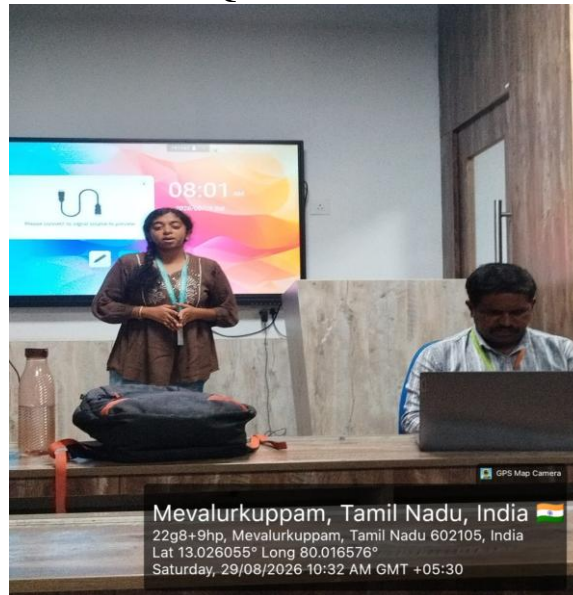
Tejaswini.M

Annexure A

Mock Interview – Sample Questions (5 Questions)

Mock Interview Test Format for Object Oriented Analysis and Design (OOAD)

Section A	Self Introduction & Fundamentals	3 Minutes	5
Section B	Technical Questions (OOAD Concepts)	7 Minutes	10
Section C	UML Modeling & Design	7 Minutes	10
Section D	Case Study / Problem Solving	8 Minutes	10
Section E	HR & Career-Oriented Questions	5 Minutes	5



SECTION A – SELF INTRODUCTION (5 Marks)

Evaluation Criteria

- Communication Skills
- Confidence
- Technical Awareness
- Academic Background

Sample Questions

1. Tell me about yourself.

Good morning sir/madam. My name is Tejaswini. I am pursuing my B.E. in Computer Science Engineering. I am interested in software development and object-oriented programming. I have worked with programming languages and technologies such as C++, Java, Python, web technologies, and databases.

2. Why did you choose Computer Science?

I chose Computer Science because I am interested in technology and problem solving. I like understanding how software works and developing solutions for real-world problems. Computer Science also gives me opportunities to work in areas such as software development, artificial intelligence, databases, and cloud technologies.

3. Explain your favorite subject in OOAD.

My favorite topic in OOAD is UML modeling, especially class diagrams and use case diagrams. A use case diagram helps us understand the interaction between users and the system, while a class diagram shows the structure of the system through classes, attributes, methods, and relationships. These diagrams make the software design easier to understand before implementation.

4. What software development tools have you used?

I have used tools such as Visual Studio Code, IntelliJ IDEA, Git, Docker, and database tools. I have also worked with programming languages such as C++, Java, Python, JavaScript, and technologies used for web application development.

5. Describe a project where you applied object-oriented concepts.

In my programming projects, I have used object-oriented concepts such as classes, objects, encapsulation, inheritance, and polymorphism. For example, in a management system, I can represent entities such as Student, Teacher, or Course as classes. Their data can be kept private using encapsulation, and common functionality can be reused using inheritance.

SECTION B – TECHNICAL QUESTIONS (15 Marks)

Basic Level

Q1. What is Object-Oriented Analysis and Design?

Answer: Object-Oriented Analysis and Design, or OOAD, is a software development approach that models a system using objects and classes. Analysis focuses on understanding what the system should do, while design focuses on how the system will implement those requirements.

Q2. Differentiate between Analysis and Design.

Answer: Analysis focuses on the problem and identifies requirements, objects, and relationships. Design focuses on the solution and defines classes, methods, interfaces, and how the objects will work together. In simple words, analysis asks what the system should do, while design asks how it will do it.

Q3. Explain Encapsulation with an example.

Answer: Encapsulation means wrapping data and methods together inside a class and controlling access to the data. For example, in a BankAccount class, balance can be private and accessed through public methods such as deposit and withdraw. It provides data protection and controlled access.

Q4. What is Inheritance? Mention its types.

Answer: Inheritance is an object-oriented concept where one class acquires properties and behavior from another class. For example, Car can inherit common features from Vehicle. The common types are single, multiple, multilevel, hierarchical, and hybrid inheritance.

Q5. Define Polymorphism and its benefits.

Answer: Polymorphism means one interface or method can have different implementations. For example, a pay method can behave differently for UPI, card, and net banking. Its benefits include flexibility, code reusability, extensibility, and easier maintenance.

Q6. What is Abstraction?

Answer: Abstraction means hiding unnecessary implementation details and showing only the essential features. For example, when we use an ATM, we perform operations such as withdrawal without knowing the internal implementation. Abstraction focuses on what an object does rather than how it does it.

Q7. Difference between Aggregation and Composition.

Answer: Aggregation is a weak has-a relationship in which the child object can exist independently. Composition is a strong has-a relationship in which the child depends on the parent for its lifetime. For example, Department and Teacher can represent aggregation, while House and Room can represent composition.

Q8. What is Coupling and Cohesion?

Answer: Coupling is the degree of dependency between different classes or modules. Cohesion is how closely related the responsibilities within one class or module are. Good software design generally aims for low coupling and high cohesion because this improves maintainability and flexibility.

Q9. What is a Class and an Object?

Answer: A class is a blueprint or template that defines attributes and methods. An object is an instance of a class. For example, Student is a class and student1 is an object of the Student class.

Q10. Explain Object Responsibility.

Answer: Object responsibility means the task or behavior that an object is responsible for performing. For example, a BankAccount object can be responsible for depositing money, withdrawing money, and checking the balance. Clear responsibilities help create focused and maintainable classes.

SECTION C – UML MODELING (15 Marks)

Q11. What is UML?

Answer: UML stands for Unified Modeling Language. It is a standard visual modeling language used to visualize, specify, design, and document software systems. Examples include Use Case, Class, Sequence, Activity, State, Component, Deployment, and Package diagrams.

Q12. When would you use a Use Case Diagram?

Answer: A Use Case Diagram is used during requirement analysis to show how external users, called actors, interact with the system. For example, in a library system, a Student can search, borrow, and return books.

Q13. Draw and explain a Class Diagram for a Library Management System.

Answer: Main classes can include Library, Book, Member, Librarian, and Issue or Transaction. A Member can borrow books, a Librarian manages books and members, and the Library contains books. The class diagram shows classes, attributes, methods, and relationships between them. Simple relationship: Library contains Book; Member borrows Book; Librarian manages Book.

Q14. Differentiate Sequence Diagram and Activity Diagram.

Answer: A Sequence Diagram shows how objects communicate with each other in a particular order over time. An Activity Diagram shows the workflow or sequence of activities in a process. Therefore, Sequence focuses on object interaction, while Activity focuses on workflow.

Q15. Explain State Diagram with an example.

Answer: A State Diagram shows the different states of an object and how it changes from one state to another because of events. For example, an order can move from Placed to Confirmed to Shipped to Delivered, or from Placed to Cancelled.

Q16. What is a Deployment Diagram?

Answer: A Deployment Diagram shows the physical arrangement of hardware and software components in a system. For example, it can show a client device, web server, application server, and database server and how they are connected.

Q17. How do Component Diagrams help software architects?

Answer: A Component Diagram shows the major software components of a system and their dependencies. It helps architects understand the high-level structure, interfaces, dependencies, and organization of large software systems.

Q18. What is Multiplicity in UML?

Answer: Multiplicity specifies how many objects of one class can be associated with an object of another class. Common values include 1, 0..1, 1..*, and 0..*. For example, one customer can place many orders, so the relationship can be one-to-many.

SECTION D – CASE STUDY / PROBLEM SOLVING (10 Marks)

Case Study 1

A hospital wants a system to manage:

- Patients
- Doctors
- Appointments
- Billing

Questions

1. Identify the classes.

2. Draw a simple class diagram.
 3. Suggest relationships among classes.
 4. Which UML diagrams would you use for analysis?
 5. Explain how inheritance can be applied.
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Case Study 2

An Online Shopping System supports:

- Customers
- Products
- Cart
- Orders
- Payments

Questions

1. Identify actors and use cases.
 2. Design a use case diagram.
 3. Suggest suitable design patterns.
 4. Explain how polymorphism can be used in payment processing.
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SECTION E – HR & PROFESSIONAL QUESTIONS (5 Marks)

Q19.Why should we hire you?

Answer: You should hire me because I am a quick learner, I have a strong interest in technology and problem solving, and I am willing to continuously improve my skills. I can understand technical concepts and apply them to practical problems. I am also comfortable working in a team and taking responsibility for my work.

Q20.What are your strengths and weaknesses?

Answer: My strengths are quick learning, problem solving, adaptability, and teamwork. One weakness I am working on is that I sometimes spend extra time trying to make my work perfect. I am improving this by setting priorities and managing my time more effectively.

Q21.How do you handle project deadlines?

Answer: First, I understand the requirements and divide the project into smaller tasks. Then I prioritize important tasks and set deadlines for each one. I regularly check my progress and

communicate with my team if there are problems. This helps me complete projects on time while maintaining quality.

Q22.Have you worked in a team project? Explain your role.

Answer: Yes, I have worked on team projects. I contribute by taking responsibility for assigned tasks, discussing technical solutions with team members, testing my work, and helping integrate different parts of the project. I believe communication and coordination are important for successful teamwork.

Q23.Where do you see yourself in five years?

Answer: In five years, I see myself as a skilled software professional with strong technical and problem-solving abilities. I want to gain experience in software development, contribute to real-world projects, continuously learn new technologies, and take greater responsibility in my organization.

RUBRICS FOR CALCULATION:**MOCK INTERVIEW**

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately